

Curvature-Aware Step-Size Adaptation and Guidance Damping for Optimal Transport Flow Matching

Generative Learning and Diffusion Models Project Proposal

July 2026

1 Goal

We investigate whether local trajectory curvature in Optimal Transport Conditional Flow Matching (OT-CFM) can serve as a trustworthy, training-free signal to co-adapt ODE integration step sizes and Classifier-Free Guidance (CFG) scales during sampling. In ideal OT-CFM, probability paths follow straight, constant-velocity trajectories ($v_t(x_t) = x_1 - x_0$) with zero acceleration ($\ddot{x}_t = 0$). However, neural network approximation errors and heavy CFG extrapolation ($w > 1$) introduce significant local trajectory curvature, bending paths off the data manifold and causing severe integration errors in low-step regimes ($N \leq 10$). We test the specific hypothesis that measuring local vector field acceleration $\mathbf{a}_t = \frac{dv_t}{dt}$ allows dynamic step-size selection and guidance damping, simultaneously reducing off-manifold artifacts and sampling steps without requiring model retraining.

2 Prior Work & Our Delta

Flow Matching (FM) [1] and Stochastic Interpolants [2] define deterministic vector fields mapping noise to data distributions along optimal transport trajectories. Fast ODE solvers such as DPM-Solver [4] leverage higher-order Taylor expansions to accelerate sampling, while Classifier-Free Guidance [3] boosts sample fidelity at the cost of high vector field curvature and trajectory distortion.

Our Delta: Existing adaptive ODE solvers rely purely on numerical truncation error estimates without considering the underlying geometry of flow matching or guidance-induced trajectory bending. We explicitly decouple trajectory curvature into velocity acceleration components, utilizing this zero-cost diagnostic signal to jointly adjust the numerical step size Δt and dynamically damp the guidance scale $w_{\text{eff}}(t)$ in high-curvature regions at test time.

3 Method

Let $x_t = (1 - t)x_0 + tx_1$ be the OT path between noise $x_0 \sim p_0$ and clean data $x_1 \sim p_1$. Under Classifier-Free Guidance (CFG), the guided velocity vector field is given by:

$$\mathbf{v}_{\text{guided}}(x_t, t, c) = \mathbf{v}_{\theta}(x_t, t, \emptyset) + w \cdot (\mathbf{v}_{\theta}(x_t, t, c) - \mathbf{v}_{\theta}(x_t, t, \emptyset)) \quad (1)$$

At inference time, we compute three dynamic operational mechanisms purely from test-time vector field evaluations:

1. **Local Trajectory Acceleration:** We estimate local acceleration via finite differences of predicted velocity vectors along the integrated trajectory:

$$\mathbf{a}_t \approx \frac{\mathbf{v}_{\text{guided}}(x_t, t + \Delta t, c) - \mathbf{v}_{\text{guided}}(x_t, t, c)}{\Delta t} \quad (2)$$

2. **Curvature-Adaptive Step Size:** Instead of a uniform grid, the step size Δt_k is updated dynamically:

$$\Delta t_k = \min \left(\Delta t_{\max}, \frac{\eta}{\|\mathbf{a}_t\|_2 + \epsilon} \right) \quad (3)$$

Where trajectory curvature is negligible ($\|\mathbf{a}_t\|_2 \approx 0$), solver steps expand to $\Delta t_{\max}$; where trajectories bend rapidly, steps contract automatically.

3. **Curvature-Damped Guidance Scaling:** To prevent extreme CFG extrapolation from driving samples off-manifold during sharp vector field transitions, we scale the effective guidance parameter:

$$w_{\text{eff}}(t) = \frac{w}{1 + \gamma \|\mathbf{a}_t\|_2} \quad (4)$$

Model & Training: We train two models: a 3-layer MLP on 2D synthetic distributions (Swiss Roll, Pinwheel) and a small U-Net ($< 5\text{M}$ parameters) on Fashion-MNIST / CIFAR-10 (32×32) using standard OT-CFM regression loss $\mathcal{L}_{\text{CFM}} = \|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2$.

Offline Evaluation Strategy: Across varying sampling budgets ($N \in \{4, 8, 12, 16\}$) and nominal guidance strengths ($w \in [1.0, 7.0]$), we evaluate generative quality using 2D Wasserstein-2 distance on synthetic data, and FID scores alongside velocity evaluation counts on image benchmarks. We compare our curvature-adaptive sampler against fixed-step Euler, Heun, and fixed-CFG baselines.

4 Resources

- **Compute:** Single workstation GPU; total training and evaluation time completes in < 1.5 hours.
- **Code & Data:** Self-contained PyTorch pipeline utilizing synthetic 2D datasets and standard image benchmarks.
- **Training:** Small models ($< 5\text{M}$ parameters) trained from scratch; no heavy pretrained weights required.

5 Challenges & Feasibility

- **Computational Overhead of Finite Differences:** Naïvely estimating $\mathbf{a}_t$ requires an additional velocity evaluation. We eliminate this overhead by caching evaluations from multi-step numerical integration steps (e.g., Heun’s predictor-corrector stages).
- **Feasibility:** Preliminary smoke tests on 2D synthetic flows confirm that acceleration spikes ($\|\mathbf{a}_t\|_2 \gg 0$) occur localized near complex manifold boundaries and high guidance scales $w > 3.0$. The lightweight parameter budget, deterministic OT paths, and fast sampling loops guarantee full execution well within standard computational constraints.

References

- [1] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow Matching for Generative Modeling,” *International Conference on Learning Representations (ICLR)*, 2023.
- [2] M. S. M. Albergo and E. Vanden-Eijnden, “Building Normalizing Flows with Stochastic Interpolants,” *International Conference on Learning Representations (ICLR)*, 2023.

- [3] J. Ho and T. Salimans, “Classifier-Free Diffusion Guidance,” *NeurIPS Workshop on Deep Generative Models*, 2021.
- [4] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.